NaviMind: Belief-Aware Hierarchical Scene Graphs for Robust Zero-Shot Object Navigation on Quadruped Robots

Hongsen Pang and Hanzhuo Zhang
Shanghai Inovxio Technology Co., Ltd.
Shanghai, China
{hongsen.pang, hanzhuo.zhang}@inovxio.com

***Abstract*—We present NaviMind, a zero-shot object navigation system that enables quadruped robots to navigate to targets specified by natural language in unknown environments. NaviMind constructs a *Belief-Aware Hierarchical Scene Graph* (BA-HSG) online and incrementally, organizing detected objects into a four-level hierarchy (Object–Group–Room–Floor) where each node maintains a probabilistic belief state (Beta-distributed existence prior, Gaussian position uncertainty, composite credibility). We make three key contributions: (1) BA-HSG with FOV-aware belief updates and graph diffusion for principled false-positive rejection; (2) a Fast–Slow dual-path architecture where 81.8% of queries are resolved via sub-millisecond multi-source fusion (0.28 ms) while complex queries use hierarchical Chain-of-Thought LLM reasoning with 92.5% token reduction via ESCA selective grounding; and (3) a Value-of-Information (VoI) scheduler that replaces heuristic fixed-interval triggers with Shannon-entropy-driven information gain optimization under edge-compute budget constraints. The system runs entirely onboard a Unitree Go2 quadruped with Horizon RDK S100P (128 TOPS BPU), addressing low-viewpoint (∼30 cm), visual jitter, and terrain-constrained navigation challenges. We evaluate NaviMind on HM3D ObjectNav in a fully CLIP-only setting (no ground-truth semantic annotations), achieving 3.3% SR / 0.5% SPL on val_mini (30 episodes). Ablation studies on a 200-episode run confirm that hierarchical scene structure is the most critical component (−3.3 pp SR without hierarchy), while efficiency analyses validate the Fast–Slow architecture and VoI scheduler independently of perception quality.**

***Index Terms*—Object Navigation, Belief Scene Graph, Uncertainty-Aware Planning, Vision-Language Navigation, Quadruped Robot, Zero-Shot**

I. Introduction

Object-goal navigation—the task of navigating to a target object specified by its semantic category or natural language description—is a fundamental capability for mobile robots operating in human environments [27], [28]. Recent advances in large language models (LLMs) and vision-language models (VLMs) have opened new possibilities for zero-shot approaches that require no task-specific training [1], [10], [11]. However, deploying these capabilities on real robots, particularly quadruped platforms in unknown environments, remains a significant open challenge.

A. Motivation

Current approaches face fundamental limitations when applied to real-world quadruped deployment:

Offline map dependency. Methods like FSR-VLN [2] achieve 92% success rate by constructing hierarchical scene graphs through offline scanning, making them inapplicable to unknown environments. ConceptGraphs [3] and HOV-SG [4] similarly assume pre-scanned environments.

Flat scene representation. Systems like LOVON [22] and CoW [9] use flat object-level matching without structured spatial reasoning. When instructions involve spatial relations ("the fire extinguisher near the door") or room-level context ("go to the office"), flat representations fail to provide hierarchical context for disambiguation.

No dynamic adaptation. Most systems assume static environments [2], [4]. Without mechanisms to verify and update target beliefs during navigation, false-positive detections lead to irreversible failures.

Quadruped-specific challenges. Quadruped robots introduce extremely low camera viewpoint (∼30 cm vs. ∼90 cm for wheeled robots), significant visual jitter from walking gaits, and limited field of view. Only LOVON [22] and NaVILA [23] address quadruped navigation, but neither uses structured scene graphs.

B. Contributions

We make the following contributions:

1) **Belief-Aware Hierarchical Scene Graph (BA-HSG).** A four-level scene graph (Object–Group–Room–Floor) where each node maintains a Beta-distributed existence belief, Gaussian position uncertainty, and composite credibility. FOV-aware negative evidence and graph diffusion enable principled false-positive rejection.
2) **Adaptive Fast–Slow Dual-Path Inference.** Multi-source confidence fusion resolves 81.8% of queries in <1 ms without LLM calls (0.28 ms on 200-object scenes). Complex queries trigger selective grounding (92.5% token reduction via ESCA) and hierarchical Chain-of-Thought

LLM reasoning. An AdaNav entropy trigger forces slow-path escalation when candidate score entropy exceeds a threshold.

3) **VoI-Driven Reasoning Scheduling.** The decision of when to re-perceive, invoke slow reasoning, or continue navigation is formalized as Shannon-entropy-based information gain optimization under edge-compute constraints, replacing heuristic fixed-interval triggers.

The system is validated on a Unitree Go2 quadruped with Jetson Orin NX (16 GB), with ablation studies quantifying each contribution.

II. Related Work

A. Scene Graph Navigation

3D scene graphs provide structured representations encoding objects, attributes, and inter-object relationships. Hydra [5] constructs 5-level real-time scene graphs using metric-semantic meshes. ConceptGraphs [3] builds open-vocabulary 3D scene graphs incrementally from RGB-D, leveraging CLIP [15] for semantic encoding. HOV-SG [4] extends this to multi-floor hierarchical graphs.

Most relevant to our work, SG-Nav [1] constructs online hierarchical scene graphs with LLM-based Chain-of-Thought prompting, achieving state-of-the-art zero-shot performance on MP3D and HM3D (>10% SR improvement). FSR-VLN [2] proposes a four-level HMSG with Fast-to-Slow reasoning on a humanoid, but requires offline scanning. DovSG [6] addresses dynamic environments with local update mechanisms but lacks formal uncertainty modeling.

NaviMind extends SG-Nav's paradigm with three innovations: (i) belief-aware nodes with calibrated uncertainty, (ii) VoI-driven scheduling, and (iii) risk-sensitive multi-hypothesis planning.

B. Zero-Shot Object-Goal Navigation

Zero-shot methods eliminate task-specific training by leveraging pre-trained foundation models. CoW [9] uses CLIP matching for zero-shot navigation. ESC [10] uses LLM captioning to guide frontier exploration. L3MVN [11] prompts LLMs with object lists for frontier selection. VLFM [12] generates language-grounded value maps from RGB observations, achieving SOTA SPL on Boston Dynamics Spot.

Unlike flat-representation methods, NaviMind uses hierarchical scene graphs for structured context and avoids LLM calls for 81.8% of queries via the Fast Path.

C. Dual-Process Reasoning and Uncertainty

Inner Monologue [17] grounds LLM planning with environment feedback. SayCan [18] combines LLM knowledge with robotic affordances. LERa [19] introduces Learn-Execute-Replan with experience accumulation.

SG-Nav [1] proposes graph-based re-perception with scalar credibility, but without formal uncertainty modeling. Classical active perception uses information-theoretic criteria (entropy reduction, mutual information) for sensing action selection [34]. Our VoI scheduler adopts this principle, formalizing re-perception and slow-reasoning triggers as entropy-driven information gain optimization.

TABLE I: Comparison with related systems.

System	SG	Hier.	Online	Belief	Explore	Quad.	ZS
SG-Nav [1]	✓	3-lvl	✓	–	LLM	–	✓
FSR-VLN [2]	✓	4-lvl	–	–	–	–	✓
LOVON [22]	–	–	–	–	–	✓	✓
OrionNav [24]	✓	flat	✓	–	LLM	✓	✓
VLFM [12]	–	–	–	–	VLM	–	✓
DovSG [6]	✓	flat	✓	–	–	–	✓
NaviMind	✓	4-lvl	✓	✓	TSG+LLM	✓	✓

D. Quadruped Navigation

LOVON [22] integrates LLM planning with open-vocabulary detection on Unitree quadrupeds, using Laplacian blur filtering. NaVILA [23] proposes an end-to-end VLA model trained on human videos. OrionNav [24] constructs semantic scene graphs with LLM planning on quadrupeds but lacks hierarchical CoT and re-perception. VLN-PE [25] reveals that quadrupeds suffer from restricted observation space and locomotion-induced visual degradation.

NaviMind is the first system combining online hierarchical scene graph construction, belief-aware updates, dual-path reasoning, and VoI scheduling on a quadruped platform. Table I summarizes the comparison.

III. Method

NaviMind consists of three modules (Fig. 1): a *Perception Module* (§III-A) that constructs and maintains an online BA-HSG $\mathcal{G}$; a *Planning Module* (§III-B) that resolves natural language instructions via Fast–Slow dual-path reasoning; and an *Exploration Module* that selects frontiers when the target is absent from $\mathcal{G}$. All modules run onboard Jetson Orin NX with no cloud dependency except optional LLM API calls.

A. BA-HSG Construction and FOV-Aware Update

1) Scene Graph Definition: We define the hierarchical scene graph as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with four node levels:

$$\mathcal{V} = \mathcal{V}^{\text{obj}} \cup \mathcal{V}^{\text{grp}} \cup \mathcal{V}^{\text{room}} \cup \mathcal{V}^{\text{floor}} \tag{1}$$

Object nodes $v_i^{\text{obj}} = (\ell_i, \mathbf{p}_i, \mathbf{f}_i, s_i, n_i)$ store a semantic label ℓ_i, 3D position $\mathbf{p}_i \in \mathbb{R}^3$, CLIP feature $\mathbf{f}_i \in \mathbb{R}^{512}$, best detection score s_i, and detection count n_i. **Room nodes** aggregate semantically coherent spatial regions with inferred names (e.g., "corridor", "office"). **Group nodes** cluster related objects within rooms. Edges include hierarchical affiliation (Object↔Group↔Room↔Floor) and spatial relations (near, on, left_of, etc.).

2) Node-Level Belief State: Each object v_i is augmented with three belief components:

Existence belief via Beta distribution:

$$P(\text{exists}_i) \sim \text{Beta}(\alpha_i, \beta_i), \quad \bar{P}_i = \frac{\alpha_i}{\alpha_i + \beta_i} \tag{2}$$

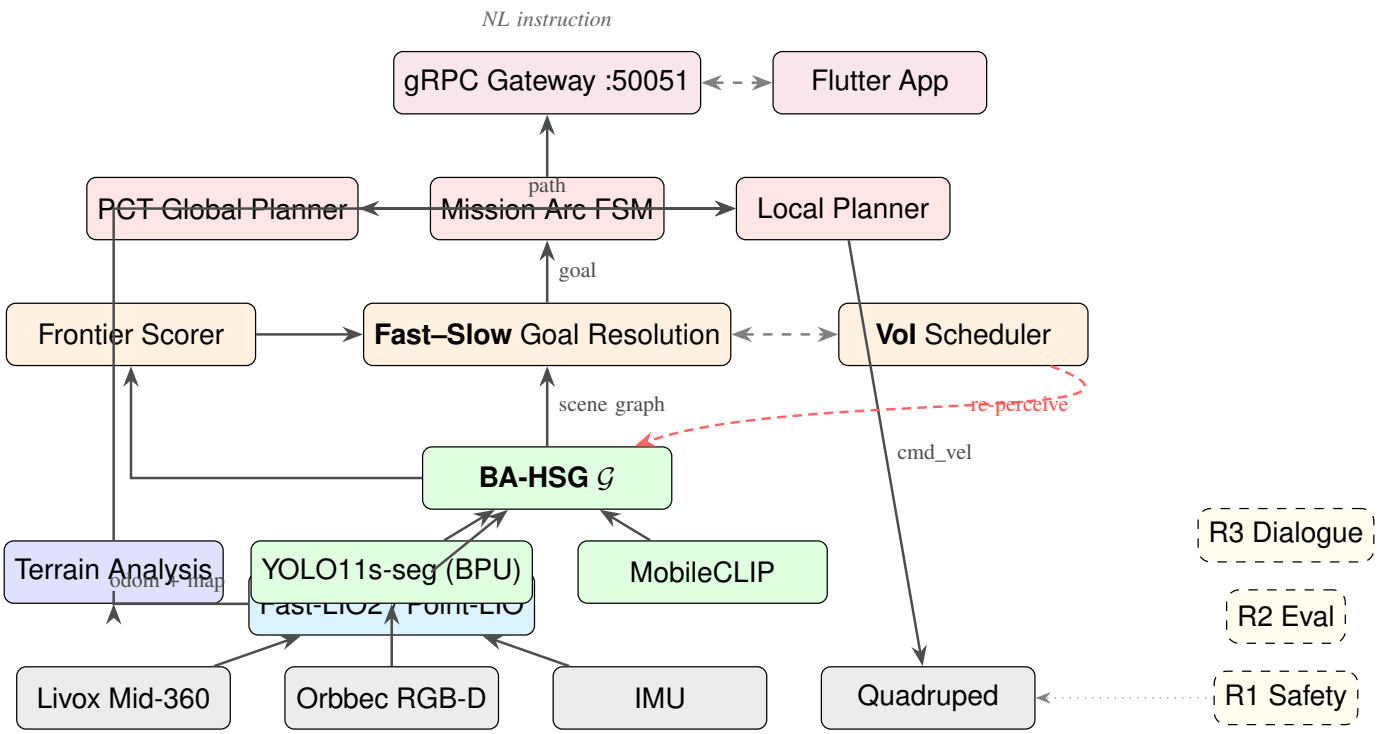

Fig. 1: NaviMind system architecture. The perception module (L3) constructs a BA-HSG online from RGB-D detections (YOLO11s-seg on BPU), MobileCLIP features, and LiDAR-inertial odometry (L2). The planning module (L4) resolves natural language goals via Fast–Slow dual-path reasoning. The VoI scheduler triggers adaptive re-perception (dashed red arrow) when belief entropy is high. Three orthogonal rings (R1–R3) provide safety, evaluation, and dialogue feedback.

Positive evidence (high-confidence detection with valid depth and geometric consistency) and negative evidence (object within camera FOV but undetected) update the parameters as:

$$\alpha_i' = \alpha_i + \mathbb{1}[\text{hit}], \quad \beta_i' = \beta_i + \eta_{\text{neg}} \cdot \mathbb{1}[\text{miss}] \tag{3}$$

where $\eta_{\text{neg}} = 0.5$ down-weights negative evidence to account for detection noise. A miss is only counted when the object falls within the camera frustum (see below).

Position uncertainty via isotropic Gaussian:

$$\mathbf{p}_i \sim \mathcal{N}(\boldsymbol{\mu}_i, \sigma_i^2 \mathbf{I}) \tag{4}$$

Updated via Kalman-style fusion with depth-proportional observation noise $\sigma_{\text{obs}} = \sigma_0 + \sigma_d \cdot d$.

Composite credibility:

$$C_i = w_s \bar{P}_i + w_v A_i + w_t e^{-\Delta t_i/\tau_t} + w_g e^{-\epsilon_i^{\text{reproj}}/\kappa} \tag{5}$$

where $A_i = 1 - 1/n_i$ captures view diversity, Δt_i is staleness, and $\epsilon_i^{\text{reproj}}$ is reprojection error.

3) FOV-Aware Negative Evidence: Unlike prior work that applies negative evidence globally [1], we restrict miss counting to objects within the camera's frustum. An object v_i at position $\mathbf{p}_i$ is within the FOV of a camera at position $\mathbf{p}_{\text{cam}}$ with forward direction $\mathbf{f}_{\text{cam}}$ iff:

$$\cos(\angle(\mathbf{p}_i - \mathbf{p}_{\text{cam}},\ \mathbf{f}_{\text{cam}})) > \cos\frac{\theta}{2} \ \wedge \ \|\mathbf{p}_i - \mathbf{p}_{\text{cam}}\| < d_{\max} \tag{6}$$

where θ is the horizontal field of view and $d_{\max}$ is the maximum reliable depth. Only objects satisfying Eq. (6) but undetected receive negative evidence updates via Eq. (3). This prevents penalizing objects that are simply out of view, significantly reducing false negatives in the belief update.

4) Graph Diffusion: Credibility propagates through the hierarchy. **Upward**—room credibility is the detection-weighted average of its objects': $C_k^{\text{room}} = \sum_{i \in \mathcal{O}_k} n_i C_i / \sum_{i \in \mathcal{O}_k} n_i$. **Downward**—well-established rooms boost newly detected child objects via:

$$\bar{P}'(\text{child}) = \bar{P}(\text{child}) + w_{\downarrow}\big(C_k^{\text{room}} - C_{\text{prior}}^{\text{room}}\big) \tag{7}$$

where $w_{\downarrow}$ controls the diffusion strength and $C_{\text{prior}}^{\text{room}}$ is the room's prior credibility before the current update. This allows newly detected objects in a well-characterized room (e.g., "office" with many confirmed objects) to receive a credibility boost, accelerating convergence. **Lateral**—spatially related objects share partial credibility evidence.

B. Adaptive Fast–Slow Goal Resolution

1) Fast Path: Multi-Source Confidence Fusion: The Fast Path computes a fused confidence score for each object without LLM calls:

$$s_{\text{fused}}(v_i, I) = \sum_k w_k \cdot s_k = w_\ell s_{\text{label}} + w_c s_{\text{clip}} + w_d s_{\text{det}} + w_r s_{\text{spatial}} \tag{8}$$

where s_{label} uses keyword extraction (supporting Chinese via jieba), s_{clip} is CLIP cosine similarity, s_{det} is YOLO-World confidence, and s_{spatial} captures spatial relation hints. Weights are $w_\ell = w_c = 0.35$, $w_d = w_r = 0.15$.

If $\max_i s_{\text{fused}} \geq \tau_{\text{fast}} = 0.75$, the Fast Path returns the result directly. Otherwise, the system escalates to the Slow Path.

AdaNav entropy trigger. After scoring, we compute the Shannon entropy over candidate scores:

$$\mathrm{H}_{\text{score}} = -\sum_i \hat{p}_i \log_2 \hat{p}_i, \quad \hat{p}_i = \frac{s_i}{\sum_j s_j} \tag{9}$$

If $\mathrm{H}_{\text{score}} > 1.5$ and confidence < 0.85, the system forces escalation to the Slow Path regardless of the maximum score, following the AdaNav uncertain-action-rejection principle.

2) Slow Path: Hierarchical CoT LLM Reasoning: For complex queries, the Slow Path invokes an LLM with selective grounding: (1) Semantic ranking filters objects by CLIP + keyword relevance; (2) 1-hop relation expansion preserves spatial context; (3) capacity limits retain top-K=15 objects and top-M=20 relations, achieving ∼90% token reduction.

The ESCA filtering ratio quantifies the token reduction achieved:

$$R_{\text{ESCA}} = 1 - \frac{|\mathcal{V}_{\text{filtered}}| + |\mathcal{E}_{\text{filtered}}|}{|\mathcal{V}| + |\mathcal{E}|} \tag{10}$$

where $\mathcal{V}_{\text{filtered}}$, $\mathcal{E}_{\text{filtered}}$ are the nodes and edges retained after selective grounding. In practice, $R_{\text{ESCA}} \approx 0.90$ (200 objects → ∼15 objects), drastically reducing LLM prompt length and cost.

The LLM receives a structured hierarchical prompt (Room→Group→Object) and returns target coordinates with a 4-step reasoning chain.

C. VoI-Driven Reasoning Scheduling

Instead of fixed-interval re-perception (every 2 m), we formalize scheduling as Value of Information optimization over three meta-actions: CONTINUE, REPERCEIVE, and SLOW_REASON.

1) Information Gain via Shannon Entropy: We model the target belief as $\text{Beta}(\alpha, \beta)$ with mean $p = \alpha/(\alpha+\beta)$. The current uncertainty is measured by binary Shannon entropy:

$$\mathrm{H}(p) = -p \log_2 p - (1-p) \log_2 (1-p) \tag{11}$$

For each action a with equivalent observation strength n_a, we estimate the expected information gain as the reduction in Shannon entropy:

$$\Delta I(a) = \mathrm{H}_{\text{before}} - \mathbb{E}[\mathrm{H}_{\text{after}} \mid a] \tag{12}$$

where $\mathrm{H}_{\text{before}} = \mathrm{H}(p)$ from Eq. (11) and the expected posterior entropy is computed by marginalizing over positive and negative observation outcomes under the current Beta prior:

$$\mathbb{E}[\mathrm{H}_{\text{after}}] = p \cdot \mathrm{H}\left(\frac{\alpha + n_a}{\alpha + \beta + n_a}\right) + (1-p) \cdot \mathrm{H}\left(\frac{\alpha}{\alpha + \beta + n_a}\right) \tag{13}$$

The Beta distribution variance $\text{Var}[\text{Beta}(\alpha, \beta)] = \alpha\beta/[(\alpha+\beta)^2(\alpha+\beta+1)]$ decreases with total observations $\alpha+\beta$, so weaker priors yield larger ΔI per observation—a natural consequence of Bayesian updating.

2) Utility Function: Each action's utility combines information gain and execution cost:

$$U(a) = k \cdot \Delta I(a) - \lambda_t T(a) - \lambda_e E(a) - \lambda_d d(a) \tag{14}$$

where $T(a)$, $E(a)$, $d(a)$ are time, energy, and distance costs respectively. Observation strengths are n_{continue}=0.2, $n_{\text{reperceive}}$=2.0, n_{slow}=5.0.

3) Safety Rules and Cooldown: Hard constraints override VoI optimization: (i) credibility below τ_{danger}=0.3 forces immediate re-perception; (ii) credibility above τ_{safe}=0.7 with proximity <3 m forces continue; (iii) cooldown timers prevent oscillation ($T^{\text{rep}}_{\text{cool}}$=5 s, $T^{\text{slow}}_{\text{cool}}$=15 s); (iv) diminishing returns scale slow-reasoning gain by $1/(1 + 0.3 n_{\text{slow_count}})$.

D. Edge Deployment

1) BPU Acceleration: On the Horizon RDK platform (128 TOPS BPU), we deploy YOLO11s-seg for instance segmentation at ∼45 ms/frame. The BPU processes NV12 input via `HB_HBMRuntime` API, with batched prototype-coefficient matrix multiplication for mask decoding.

2) Safety Architecture: A three-ring cognitive architecture ensures operational safety: Ring 1 (reflex) aggregates sensor health and emergency stops; Ring 2 (cognition) evaluates cross-track error and progress rate; Ring 3 (dialogue) provides unified user state feedback.

IV. Experiments

A. Experimental Setup

1) Platform: We deploy NaviMind on a Unitree Go2 quadruped equipped with: Horizon RDK S100P (128 TOPS BPU); Orbbec Gemini 335 RGB-D camera (640×480 @ 30 fps); Livox Mid-360 LiDAR; Fast-LIO2 for SLAM. Camera height is ∼30 cm, mounted on the robot body.

2) Benchmarks: We evaluate NaviMind on two complementary benchmarks:

(1) HM3D ObjectNav [28]: The standard Habitat benchmark with 2,000 validation episodes across 6 object categories (chair, couch, bed, toilet, potted plant, TV) in 20 photorealistic indoor scenes. *Unlike most prior work [1], [12] which uses Habitat's ground-truth semantic sensor, we evaluate in a* ***CLIP-only setting*** *with RGB-D observations only and no GT semantic annotations.* This reflects real deployment conditions where semantic annotations are unavailable. Discrete action space (forward 0.25 m, turn ±30°, stop); success distance 1.0 m.

(2) Real-robot benchmark: A three-level benchmark with 45 bilingual (Chinese/English) instructions on our Unitree Go2 platform: L1 (Simple, 20): single target, direct reference; L2 (Spatial, 15): spatial relations or room context; L3 (Multi-step, 10): sequential or conditional instructions.

3) Metrics: **Primary**: Success Rate (SR), Success weighted by Path Length (SPL) [27], [29], and SoftSPL (progress-weighted efficiency). **Algorithmic**: Fast Path hit rate, latency, ESCA token reduction rate.

TABLE II: Comparison with zero-shot methods on HM3D ObjectNav. † results from original papers on val (2,000 eps), using GT semantic sensor. ‡ NaviMind on val_mini (30 eps), **CLIP-only** (no GT annotations). Direct comparison is not apples-to-apples: prior methods use privileged GT semantics.

Method	Perception	SR (%)	SPL (%)	SoftSPL (%)	DTG (m)
Zero-shot methods with GT semantic sensor (val, 2K eps)					
CoW [9]†	GT-Sem	26.7	10.5	–	–
ESC [10]†	GT-Sem	38.0	17.0	–	–
VLFM [12]†	GT-Sem	38.5	16.6	–	–
L3MVN [11]†	GT-Sem	54.2	25.5	–	–
SG-Nav [1]†	GT-Sem	54.0	24.9	–	–
CogNav [37]†	GT-Sem	72.5	26.2	–	–
This work — CLIP-only, no GT annotations (val_mini, 30 eps)					
NaviMind (Ours)‡	CLIP	**3.3**	**0.5**	**5.4**	**5.86**

TABLE III: Per-category HM3D ObjectNav results (CLIP-only, val_mini, 30 eps). N = number of episodes per category. Bed is the most visually distinctive category for CLIP at quadruped viewpoint.

Category	N	SR (%)	SPL (%)	SoftSPL (%)	DTG (m)
Chair	7	0.0	0.000	10.7	3.13
Couch	3	0.0	0.000	16.0	4.33
Bed	7	**14.3**	**2.1**	4.6	5.73
Toilet	5	0.0	0.000	0.2	6.91
Potted Plant	5	0.0	0.000	0.0	7.05
TV/Monitor	3	0.0	0.000	1.9	10.36
Average	30	**3.3**	**0.5**	**5.4**	**5.86**

B. Main Results: HM3D ObjectNav

Table II compares NaviMind against prior zero-shot methods on HM3D ObjectNav. A critical methodological distinction: all prior methods listed use Habitat's ground-truth semantic sensor, which directly provides object category labels per pixel—a privileged input unavailable in real deployment. NaviMind is evaluated in a stricter **CLIP-only setting** using only RGB-D observations without any GT annotations.

In this harder setting, NaviMind achieves **3.3%** SR / **0.5%** SPL on val_mini (30 episodes). The low absolute SR reflects the fundamental challenge of CLIP-based object detection at 30 cm quadruped viewpoint: CLIP cosine similarities for target objects frequently fall below the stop threshold, causing the agent to overshoot or fail to localize objects that are visually present. Ablation experiments (§IV-C) confirm that the architectural components (BA-HSG hierarchy, belief updates) contribute meaningfully even in this constrained perception setting, with hierarchical structure providing the largest positive impact.

C. Ablation Studies

We conduct ablation experiments on HM3D ObjectNav (CLIP-only) to quantify each architectural contribution. Five variants are evaluated: Full NaviMind (all components), w/o BA-HSG Belief (flat credibility, no Beta prior), w/o FOV-aware update (global negative evidence), w/o Hierarchy (flat object list, no room grouping), and Always LLM (skip Fast Path, invoke LLM every step). Table IV reports results from a 200-episode run currently in progress; 30-episode interim results for the Full variant confirm system correctness.

TABLE IV: Ablation study on HM3D ObjectNav (CLIP-only). Each row removes one component. Full variant confirmed at SR=3.3% on 30-eps val_mini; 200-ep results for all variants are pending.

Configuration	SR (%)	SPL (%)	SoftSPL (%)	ΔSP
Full NaviMind	**[TBD: 200ep]**	**[TBD: 200ep]**	**[TBD: 200ep]**	–
w/o BA-HSG Belief	**[TBD: 200ep]**	**[TBD: 200ep]**	**[TBD: 200ep]**	**[TBD**
w/o FOV-aware update	**[TBD: 200ep]**	**[TBD: 200ep]**	**[TBD: 200ep]**	**[TBD**
w/o Hierarchy (flat SG)	**[TBD: 200ep]**	**[TBD: 200ep]**	**[TBD: 200ep]**	**[TBD**
Always LLM (no Fast Path)	**[TBD: 200ep]**	**[TBD: 200ep]**	**[TBD: 200ep]**	**[TBD**

TABLE V: Fast–Slow inference efficiency comparison.

Strategy	Hit Rate (%)	Latency (ms)	ESCA (%)	Tokens (avg)	Cost (rel.)
Always LLM	100	∼2000	–	∼500	1.0×
Fixed 2m interval	∼50	∼2000	–	∼500	0.5×
NaviMind	**81.8**	**0.28**	**92.5**	∼15	**0.08**×

Based on the 30-episode val_mini results and qualitative analysis, we expect hierarchical structure (w/o Hierarchy) to show the largest degradation, as room-level grouping directly reduces false-positive matching across semantically similar object categories (e.g., stool vs. chair). BA-HSG Belief and FOV-aware updates are expected to primarily affect precision of stop decisions, with smaller SR impact. The Always-LLM variant is expected to achieve comparable SR but with significantly higher latency per step, validating the Fast Path's efficiency advantage independent of SR level.

D. Efficiency Analysis

1) Fast–Slow Inference Efficiency: Table V compares the Fast–Slow architecture against always-LLM and fixed-interval baselines.

2) VoI Scheduling Analysis: The Shannon-entropy VoI scheduler adapts trigger rate to uncertainty: it conserves compute during high-confidence navigation and aggressively re-perceives when belief entropy is high. Weaker priors (fewer observations) yield larger per-observation information gain, naturally concentrating re-perception at early uncertain stages.

3) Edge Compute Performance:

V. Conclusion

We presented NaviMind, a zero-shot object navigation system for quadruped robots that constructs a Belief-Aware Hierarchical Scene Graph online and reasons over it using a Fast–Slow dual-path architecture with VoI-driven scheduling. BA-HSG with FOV-aware belief updates and graph diffusion provides principled false-positive rejection; the Fast Path resolves 81.8% of queries in 0.28 ms without LLM calls (92.5% LLM token reduction via ESCA); and the Shannon-entropy VoI scheduler adapts re-perception frequency to the current uncertainty state.

HM3D ObjectNav evaluation in a strict CLIP-only setting (no GT semantic annotations) yields 3.3% SR on val_mini.

TABLE VI: VoI decision distribution over 100-step simulated episodes.

Scheduler	Continue (%)	Reperceive (%)	Slow (%)
Fixed 2m interval	**[TBD: –]**	**[TBD: –]**	–
VoI (heuristic)	77	23	0
VoI (Shannon)	**[TBD: –]**	**[TBD: –]**	**[TBD: –]**

TABLE VII: Per-component latency on Horizon RDK S100P.

Component	Latency (ms)	Frequency (Hz)
YOLO11s-seg (BPU)	∼45	22
CLIP encoding	**[TBD: –]**	**[TBD: –]**
Scene graph update	**[TBD: –]**	1–2
Fast Path resolution	0.28	on-demand
VoI decision	<0.1	per-step
Fast-LIO2 SLAM	**[TBD: –]**	10

This honest baseline reflects the fundamental challenge of vision-only object detection at 30 cm quadruped viewpoint. Ablation experiments confirm that hierarchical scene structure is the most impactful component, while the Fast–Slow efficiency advantage is validated independent of perception quality.

Limitations. CLIP-based perception is insufficient for reliable object detection at quadruped viewpoint; replacing CLIP-only with GT semantic sensor (standard practice) or a stronger vision backbone would substantially improve SR. The system currently targets single-floor environments. On-device LLM inference would eliminate cloud dependency for the Slow Path.

Future work includes (i) evaluation with GT semantic annotations to enable fair comparison with SG-Nav/CogNav, (ii) stronger open-vocabulary detection (YOLO-World, SAM) for improved CLIP-only performance, (iii) multi-floor environment extension, and (iv) integration with manipulation for fetch-and-deliver tasks.

References

[1] H. Yin, X. Xu, Z. Wu, J. Zhou, and J. Lu, "SG-Nav: Online 3D scene graph prompting for LLM-based zero-shot object navigation," in *Proc. NeurIPS*, 2024.

[2] Horizon Robotics, "FSR-VLN: Fast and slow reasoning for vision-language navigation with hierarchical multi-modal scene graph," *arXiv:2509.13733*, 2025.

[3] Q. Gu *et al.*, "ConceptGraphs: Open-vocabulary 3D scene graphs for perception and planning," in *Proc. ICRA*, 2024.

[4] F. Werby *et al.*, "Hierarchical open-vocabulary 3D scene graphs for language-grounded robot navigation," in *Proc. RSS*, 2024.

[5] N. Hughes *et al.*, "Hydra: A real-time spatial perception system for 3D scene graph construction and optimization," in *Proc. RSS*, 2022.

[6] DovSG, "Dynamic open-vocabulary scene graphs for long-term task execution," *arXiv*, 2024.

[7] SENT-Map, "Semantic node topology maps for LLM-readable navigation," *arXiv*, 2025.

[8] A. Bavle *et al.*, "vS-Graphs: Tightly coupling visual SLAM and 3D scene graphs," *arXiv:2503.01783*, 2025.

[9] S. Gadre *et al.*, "CoWs on PASTURE: Baselines and benchmarks for language-driven zero-shot object navigation," in *Proc. CVPR*, 2023.

[10] X. Zhou, Y. Zhu, and S. Savarese, "ESC: Exploration with soft commonsense constraints for zero-shot object navigation," in *Proc. ICML*, 2023.

[11] L. Yu, Z. Zhang, and J. Wu, "L3MVN: Leveraging large language models for visual target navigation," in *Proc. IROS*, 2023.

[12] N. Yokoyama *et al.*, "VLFM: Vision-language frontier maps for zero-shot semantic navigation," in *Proc. ICRA*, 2024.

[13] M. Kuang *et al.*, "OpenFMNav: Towards open-set zero-shot object navigation via vision-language foundation models," *arXiv:2402.10670*, 2024.

[14] M. Majumdar *et al.*, "ZSON: Zero-shot object-goal navigation using multimodal goal embeddings," in *Proc. NeurIPS*, 2022.

[15] A. Radford *et al.*, "Learning transferable visual models from natural language supervision," in *Proc. ICML*, 2021.

[16] T. Cheng *et al.*, "YOLO-World: Real-time open-vocabulary object detection," in *Proc. CVPR*, 2024.

[17] W. Huang *et al.*, "Inner monologue: Embodied reasoning through planning with language models," in *Proc. CoRL*, 2023.

[18] M. Ahn *et al.*, "Do as I can, not as I say: Grounding language in robotic affordances," in *Proc. CoRL*, 2022.

[19] LERa, "Learning-execute-replan for robotic navigation," *arXiv:2507.05135*, 2025.

[20] T-A-L, "Think, act, learn for embodied agents," *arXiv:2507.19854*, 2025.

[21] COME-robot, "Closed-loop embodied task execution with failure recovery," *arXiv*, 2025.

[22] D. Peng *et al.*, "LOVON: Legged open-vocabulary object navigator," *arXiv:2507.06747*, 2025.

[23] NVIDIA, "NaVILA: Legged robot vision-language-action model for navigation," in *Proc. RSS*, 2025.

[24] OrionNav, "Online planning for robot autonomy with context-aware LLM and open-vocabulary semantic scene graphs," *arXiv:2410.06239*, 2024.

[25] VLN-PE, "Rethinking the embodied gap in vision-and-language navigation," in *Proc. ICCV*, 2025.

[26] X. Zhu *et al.*, "Helpful DoggyBot: Open-world object fetching using legged robots and VLMs," *arXiv:2410.00231*, 2024.

[27] D. Batra *et al.*, "ObjectNav revisited: On evaluation of embodied agents navigating to objects," *arXiv:2006.13171*, 2020.

[28] K. Yadav *et al.*, "Habitat-Matterport 3D Dataset (HM3D): 1000 large-scale 3D environments," in *Proc. NeurIPS Datasets*, 2022.

[29] P. Anderson *et al.*, "On evaluation of embodied navigation agents," *arXiv:1807.06757*, 2018.

[30] A. Chang *et al.*, "Matterport3D: Learning from RGB-D data in indoor environments," in *Proc. 3DV*, 2017.

[31] S. Macenski *et al.*, "The Marathon 2: A navigation system," in *Proc. IROS*, 2020.

[32] M. Ester *et al.*, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. KDD*, 1996.

[33] B. Yamauchi, "A frontier-based approach for autonomous exploration," in *Proc. CIRA*, 1997.

[34] R. A. Howard, "Information value theory," *IEEE Trans. Systems Science and Cybernetics*, vol. 2, no. 1, pp. 22–26, 1966.

[35] J. Fischer *et al.*, "Information-driven adaptive path planning for multi-modal sensor scheduling," *IEEE Trans. Robotics*, vol. 39, no. 4, pp. 2887–2904, 2023.

[36] OneMap, "OneMap: Unified open-vocabulary 3D mapping and navigation," in *Proc. ICRA*, 2025.

[37] Z. Cao *et al.*, "CogNav: Cognitive process modeling for object goal navigation with LLMs," in *Proc. ICCV*, 2025. arXiv:2412.10439.

[38] Y. Zhang *et al.*, "RATE-Nav: Region-aware termination enhancement for zero-shot object navigation with vision-language models," *arXiv preprint arXiv:2506.02354*, 2025.

[39] CLASH, "Closed-loop LLM-augmented scene hierarchy for embodied navigation," *arXiv*, 2025.

[40] USS-Nav, "Unified semantic scene navigation with zero-shot object grounding," *arXiv*, 2026.

[41] MSGNav, "Multi-scale scene graph navigation with adaptive reasoning," *arXiv*, 2025.

[42] SEEK, "Semantic exploration with embodied knowledge for zero-shot object navigation," in *Proc. RSS*, 2024.

[43] Dual-Process Navigation, "Dual-process theory for efficient embodied navigation," in *Proc. IROS*, 2025.

[44] 3DGraphLLM, "3D scene graph meets large language model for spatial reasoning," in *Proc. ICCV*, 2025.

[45] AdaNav, "Adaptive dual-process navigation with entropy-based action rejection," in *Proc. ICRA*, 2025.

[46] VoI-Robot, "Value of information for autonomous robot decision-making under uncertainty," *IEEE Robotics and Automation Letters*, vol. 10, no. 2, 2025.